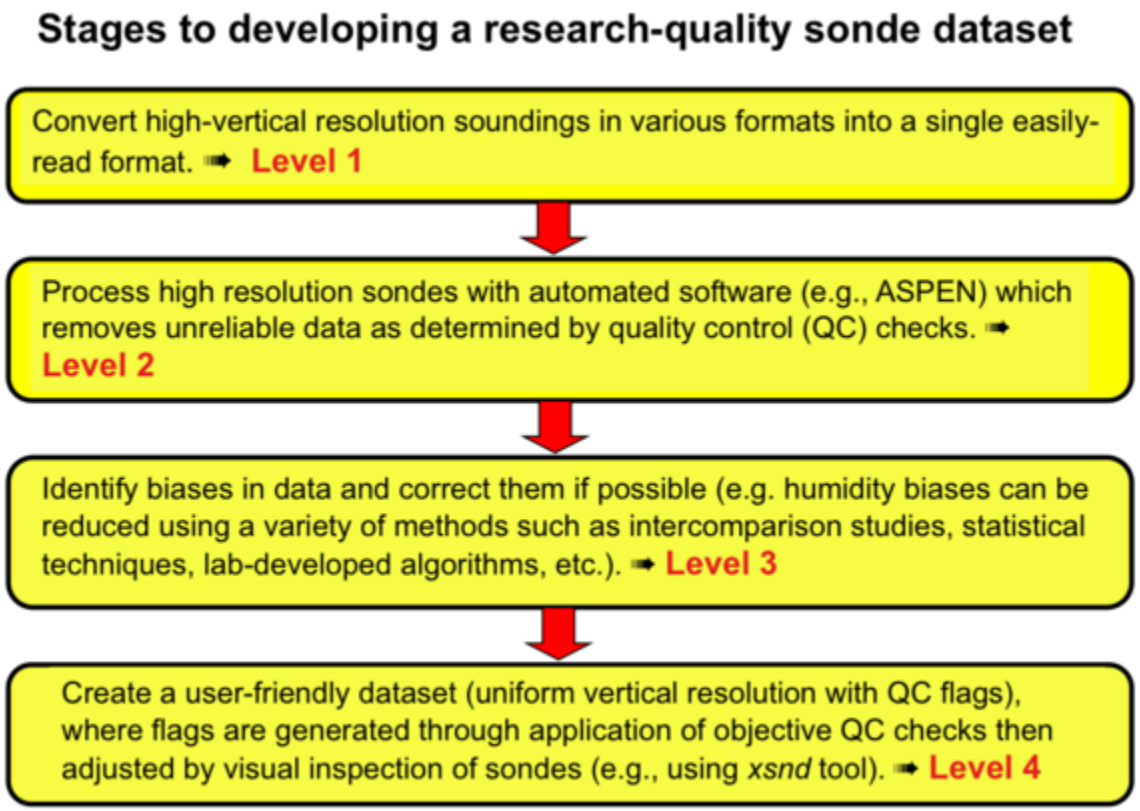
CDF-based Correction of Storm Tracker

Workflow and code

2024-09-19

Workflow

- Collect raw data
- **From level 0 to level 1**: format unification.
- **From level 1 to level 2**: correct known errors such as missing values and impossible data.
- Perform **CDF-based correction** (and generate level 3 data)



Developing High-Quality Field Program Sounding Datasets (Ciesielski et al., 2011)

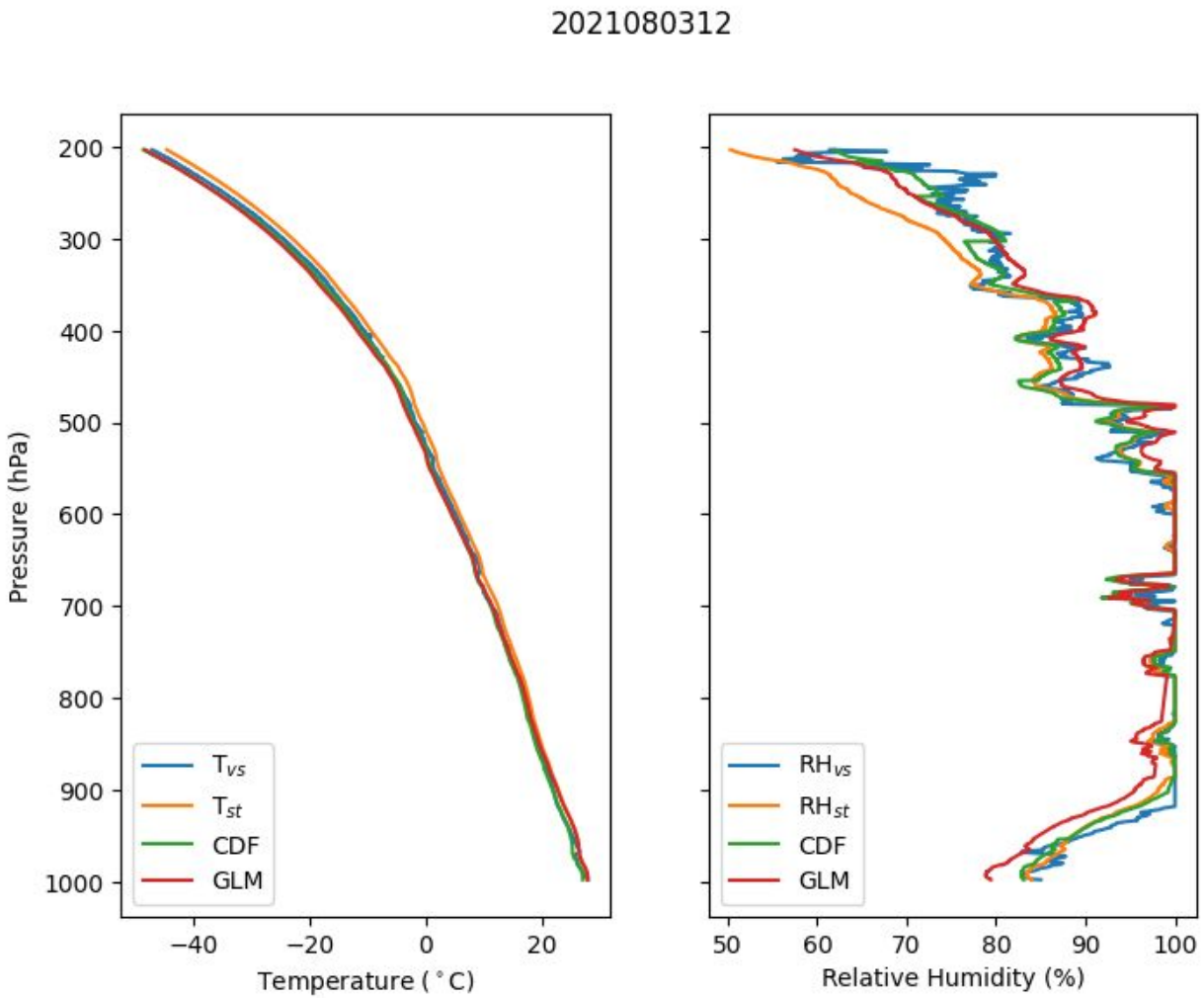
Example: 2021-0803-12Z

ST_L0_2021080312_3014.csv

ST_L1_2021080312_3014.csv

ST_L2_2021080312_3014.csv

ST_L2_2021080312_3014_cdf_corrected.csv



Prerequisites of CDF-based Correction

- Python (free)
  - You may download Python through the official website or Anaconda (which provides pre-installed packages for various purposes).
- Python packages used (free):
  - NumPy: the fundamental package for scientific computing with Python
  - Pandas: pandas is a fast, powerful, flexible, and easy-to-use open-source data analysis and manipulation tool.
- Note: the two packages are pre-installed if you use Anaconda. Otherwise, you may install it with the commands:

```
> pip install numpy pandas
```

Before Using CDF-based Correction

- **Preprocess** your ST observation data to **level 2**.
- **Define the timestamp** of the observation in YYYYMMDDHH (in UTC). For example, **2024091902** is the timestamp for our online meeting (2024-09-19 10:00 AM Taipei Time).
- **Define dp0**, the ground check pressure reference. This value is the difference between the pressure measure by ST at launch (P_st_0) and the reference pressure sensor (P_0).

$$dp0 = P_st_0 - P_0$$

Using CDF-based Correction

- In the directory of the code and model files:
  - perfrom_cdf_correction.py
  - cdf_T_correction.csv
  - cdf_RH_correction.csv
- Use command:

```
> python perform_cdf_correction.py -i ST_L2_2021080312_3014.csv -t 2021080312 --dp0 1.5941
```

- Results in ST_L2_2021080312_3014_cdf_corrected.csv and some statistics.

```
python perform_cdf_correction.py -h
```

- You may type for more options.